

# TATA POWER

## EV CHARGING DATA ANALYSIS

Project Documentation & Business Insights

|                   |                                                                                 |
|-------------------|---------------------------------------------------------------------------------|
| Focus             | Network performance, station utilization, customer demand and financial returns |
| Analytical themes | Performance • Profitability • Expansion                                         |
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This documentation is prepared from the supplied Tata Power EV Charging project report and its dashboard views. It organizes the reported findings into a portfolio-ready business analysis document.

# 1. Executive Summary

A concise view of the business question, reported findings and decision areas.

The project evaluates EV charging network performance by combining operational usage, customer behaviour and financial-return indicators. The report frames the strategic mandate around three areas: improving station performance, evaluating profitability and directing future expansion.

## Performance

Assess station utilization, charging volume and energy-consumption patterns to identify areas where actual demand differs materially from available infrastructure.

## Profitability

Use revenue, payback and EBITDA-oriented indicators to understand portfolio health and the economics of infrastructure investment.

## Expansion

Identify high-demand districts and suitable site types so future charging infrastructure can be aligned with observed demand and reported ROI metrics.

## Headline reported findings

- The report highlights Jayanagar and Rajajinagar as major demand centres with substantial infrastructure deficits.
- The report identifies sharp usage drops on Fridays and Saturdays, creating an opportunity for demand-generation tactics.
- Mall/retail installations are reported as having the longest average payback period, while residential-complex installations are reported with a shorter payback period.
- The financial dashboard reports a portfolio ROI of 41.43% and average payback of 5 years.

## 2. Business Mandate & Analytical Objectives

The supplied report describes the objective as combining operational data, customer usage and financial ROI metrics to guide investment decisions, optimize station performance and identify high-potential locations for future expansion.

### Key business questions

- Where is charging demand concentrated, and where does actual demand exceed projected infrastructure capacity?
- How efficiently are charging stations and plugs being utilized?
- What temporal patterns should be considered when planning operations and demand-generation activities?
- Which customer and vehicle segments contribute to charging demand and revenue?
- How do payback, ROI and EBITDA-oriented measures inform expansion decisions?

### Analytical framework

| Performance                                                  | Profitability                       | Expansion                                                   |
|--------------------------------------------------------------|-------------------------------------|-------------------------------------------------------------|
| Station utilization<br>Energy consumption<br>District demand | Revenue<br>ROI<br>Payback<br>EBITDA | District targeting<br>Site selection<br>Capacity allocation |

### Portfolio use

For a Data Analyst portfolio, this project demonstrates how dashboard metrics can be translated into business questions, diagnosed operational gaps and actionable recommendations.

### 3. KPI & Dashboard Overview

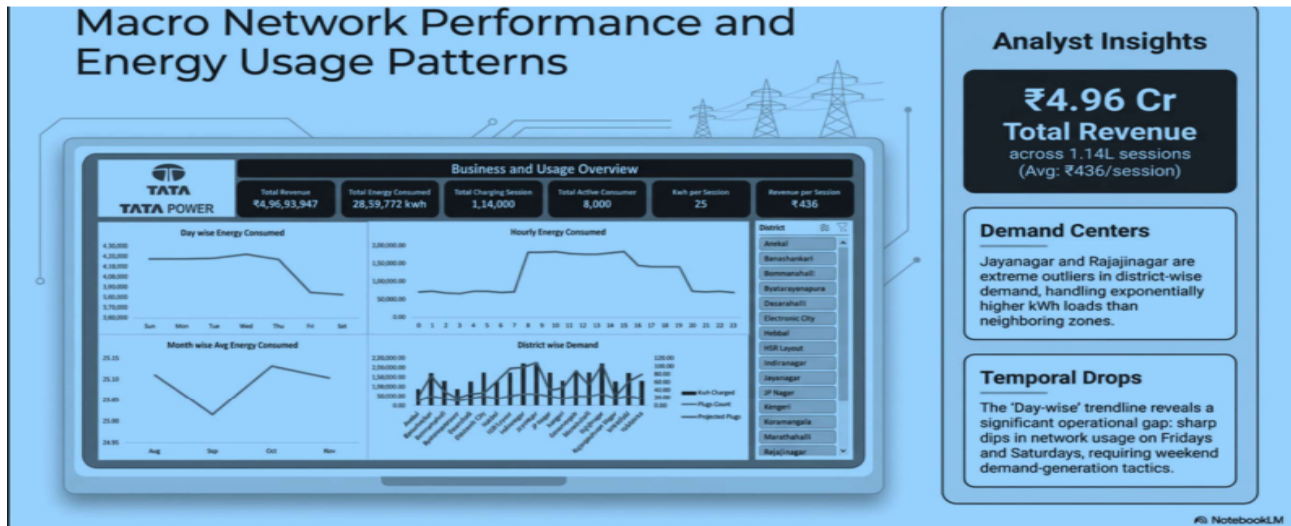
Reported dashboard measures visible in the supplied project report.

| KPI                 | Reported value       | Context                                       |
|---------------------|----------------------|-----------------------------------------------|
| Total Revenue       | ₹4.96 Cr             | Reported in the network performance view.     |
| Charging Sessions   | 1.14 million         | Reported total session volume.                |
| Revenue / Session   | ₹436                 | Reported average revenue per session.         |
| Active Customers    | 8,000                | Reported in the business and usage dashboard. |
| Average Utilization | 292.31 kWh / station | Reported system average.                      |
| Sessions / Plug     | 7,333                | Reported system average.                      |
| Portfolio ROI       | 41.43%               | Reported in the financial performance view.   |
| Average Payback     | 5 years              | Reported in the financial performance view.   |

Note: Values above are transcribed from the supplied dashboard/report visuals. The source report is primarily visual, so this documentation does not add unsupported raw-data definitions or methodology details.

## 4. Network Performance & Energy Usage

The network-performance dashboard compares revenue, energy consumed, charging sessions and active customers while also showing day-wise, hourly, monthly and district-wise demand patterns.

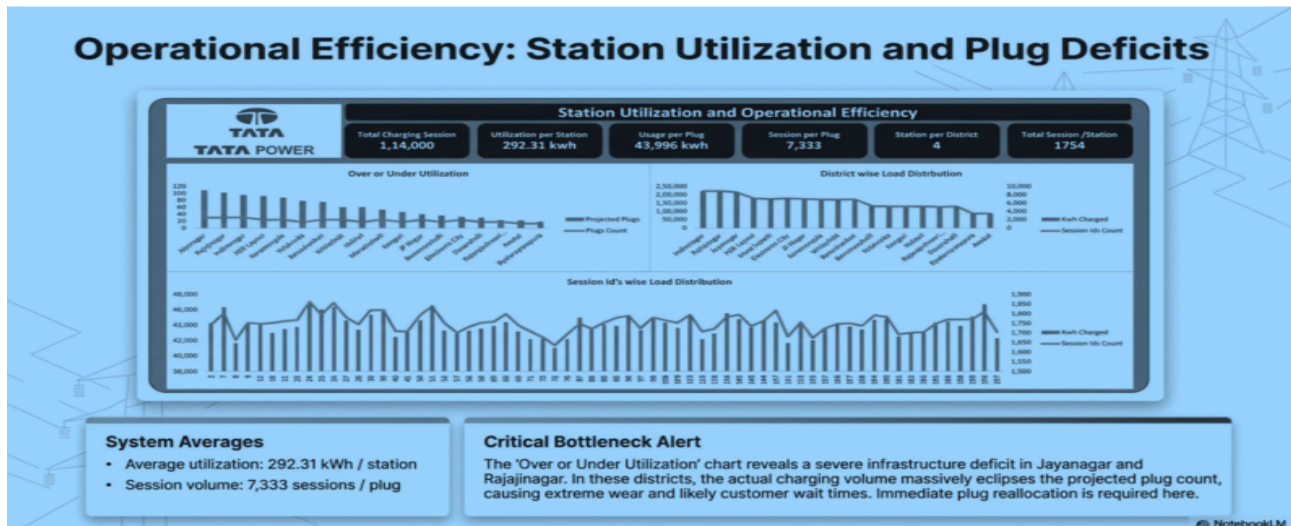


### Reported observations

- Jayanagar and Rajajinagar are highlighted as extreme demand centres in district-wise demand.
- The report states that these districts handle substantially higher kWh loads than neighbouring zones.
- The day-wise trend shows noticeable drops in network usage on Fridays and Saturdays.

## 5. Operational Efficiency & Station Utilization

The operational dashboard focuses on station-level utilization, projected versus actual plug demand, district-wise load distribution and session-level energy usage.

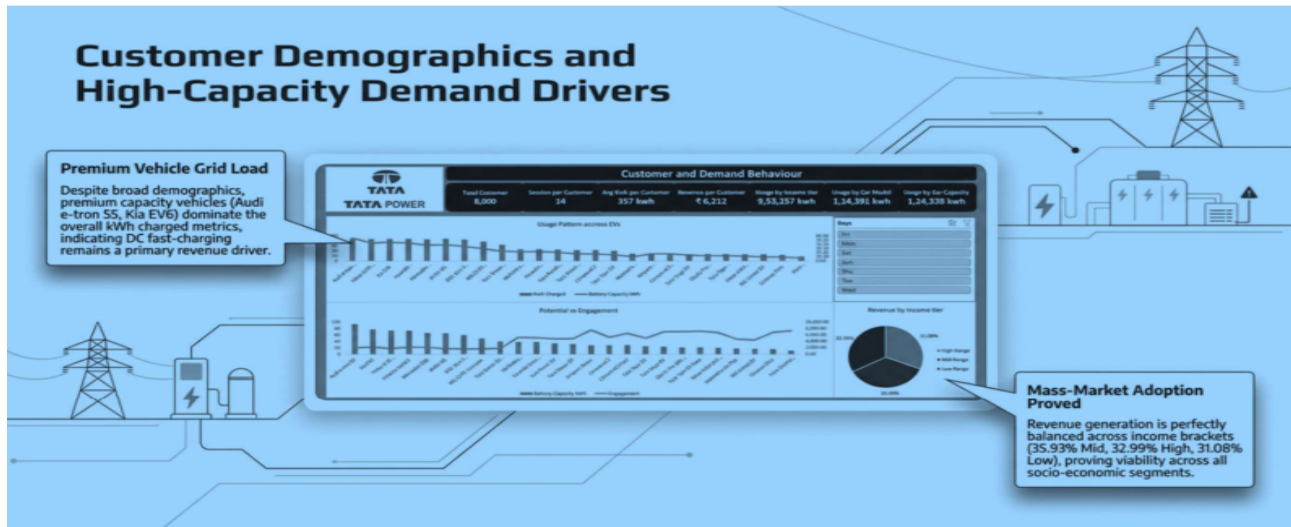


### Reported operational insight

- The system average is reported as 292.31 kWh per station.
- The dashboard reports 7,333 sessions per plug.
- The report identifies a severe infrastructure deficit in Jayanagar and Rajajinagar where actual charging volume exceeds projected plug capacity.
- The recommended response in the source report is to reallocate infrastructure toward these high-demand districts.

## 6. Customer Demographics & High-Capacity Demand Drivers

The customer-behaviour dashboard examines customer count, sessions per customer, energy usage and charging-capacity patterns, along with revenue by income segment.

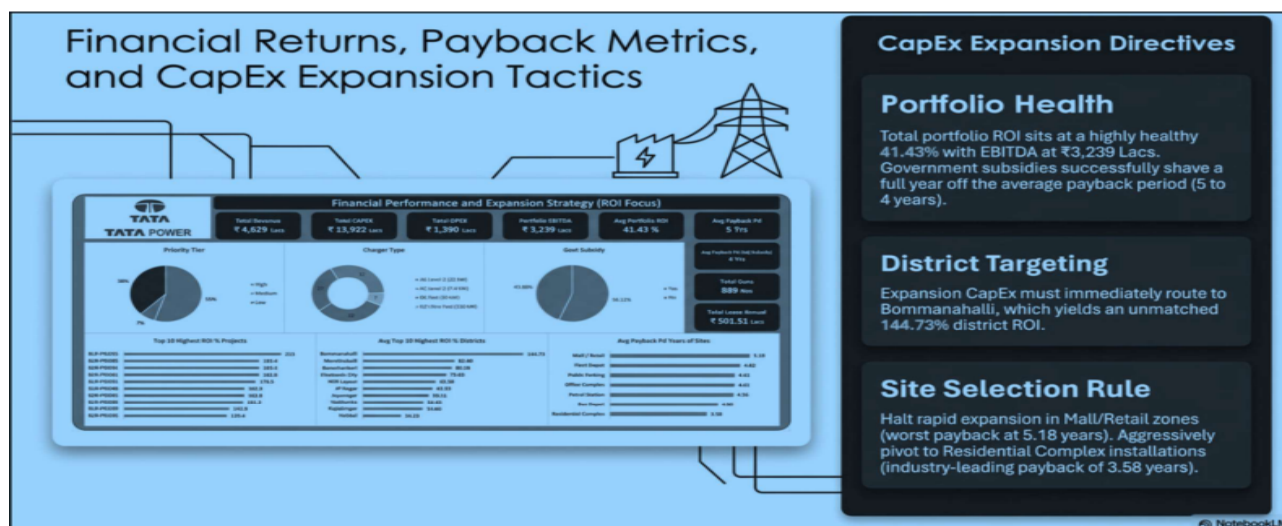


### Reported customer insights

- The report describes premium vehicle groups as an important contributor to high-charged-energy demand.
- The report also states that revenue is distributed across income brackets, with the mass-market segment contributing materially to the overall revenue mix.
- The dashboard connects battery/energy capacity with usage patterns to identify higher-capacity demand drivers.

## 7. Financial Returns & CapEx Expansion

The financial dashboard evaluates revenue, total CapEx, ROI, EBITDA-oriented measures, project ranking and payback period. It is intended to support portfolio-level expansion decisions.



### Reported financial findings

- Portfolio ROI is reported at 41.43%.
- Average payback is reported at 5 years.
- The report states that Mall/Retail sites have the longest average payback at 5.18 years.
- Residential Complex installations are reported with a 3.58-year payback in the strategic recommendations.
- The report highlights district targeting, portfolio health and site-selection rules as the three major CapEx decision areas.

## 8. Strategic Recommendations

| Area              | Recommendation from the project                                                                  | Business rationale                                                                                |
|-------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Infrastructure    | Reinforce infrastructure in Jayanagar and Rajajinagar.                                           | These districts are reported as high-demand areas with a plug-capacity deficit.                   |
| Expansion mix     | Shift future expansion toward Residential Complexes and high-ROI districts such as Bommanahalli. | The source report links these locations/site types with shorter payback or stronger reported ROI. |
| Demand generation | Consider time-of-use pricing or promotional tactics for low-demand periods.                      | The report identifies Friday/Saturday usage drops as an operational opportunity.                  |
| Site selection    | Use payback, ROI and demand together when prioritizing new sites.                                | This keeps expansion decisions connected to both operational demand and financial returns.        |

### Analyst takeaway

The project demonstrates a business-analysis workflow in which operational demand, customer usage and financial metrics are viewed together. The central decision theme is to align infrastructure capacity and future CapEx with observed demand and reported return measures.

## 9. Conclusion & Portfolio Structure

The Tata Power EV Charging project converts charging-network data into a set of operational and financial insights. The supplied report emphasizes demand concentration, station utilization, customer behaviour, ROI and payback as the main decision dimensions.

### Suggested GitHub repository structure

| File                             | Purpose                                                                     |
|----------------------------------|-----------------------------------------------------------------------------|
| Tata_Power_Project_Report.pdf    | Original project report/presentation.                                       |
| Tata_Power_EV_Charging_Data.xlsx | Source/working data file, if included in the project.                       |
| Dashboard_Screenshot.png         | Representative Power BI/dashboard screenshot.                               |
| Project_Documentation.pdf        | Business-focused project documentation and insights.                        |
| README.md                        | Short project overview, tools, objectives, key findings and links to files. |

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Source basis: Tata Power EV Charging project report supplied in this conversation.